Update of the aircraft model versus the main paper

Comparison of the pre-computed optimization results, organised like the paper (Methods, then Results)

Cumulative CO_2 is expressed as a percentage of the 3 % share of the 2.0 °C (66 %) carbon budget allocated to aviation, as in the paper; 'paper → update' always reads main paper result → update result. All numbers are read from the committed optima, none was recomputed by hand.

Summary

The update model (`aircraft_model="update"`) replaces GAM V2.0 by GAM V3.0 and adds powertrain scale laws with a maximum power per propulsor, a revised fuel cell system with thermal management, a size-dependent LH2 tank gravimetric index, an engine efficiency trend and four-point technology scenarios (2020, 2040, 2060, 2080). Energy pathways, background scenarios, demand model, controls and optimizer are those of the main paper, and the timeline is the paper's (2025–2075, entry into service up to 2060). Entry into service is bounded below by 2035 (Jet-A) and 2040 (alternative architectures) in the update, and the LH2 gas turbine and fuel cell start at the same date.

- **Emissions are higher in every scenario.** For the Mid technology breakthrough scenarios, the cumulative emissions go from 166 to 216 % of the budget (trend), 130 to 177 % (availability); the increase ranges from 37 to 89 points of budget over all scenarios without demand reduction and technology levels.
- **The budget is respected only with demand reduction.** The low-demand variants stay feasible in both models (100 % of the budget) but need more avoided demand in the update (39 → 45 % of the trend traffic in 2075).
- **The ranking of the scenarios is unchanged**: baseline > drop-in > breakthrough, availability and low demand reduce emissions, higher technology reduces emissions.
- **The architectures change.** Fuel cell aircraft lose share (Mid technology: 57 → 29 % of the 2075 supplied ASK in the breakthrough trend scenario, 85 → 44 % with availability; 30–57 % in the paper's Lower scenario, none in the update). The LH2 gas turbine, the Jet-A new generation (v2) and the battery-electric aircraft gain share (see Part II, Fleet mix).
- **Why:** the small-turbofan penalty of GAM V3.0 and its refitted regressions make the general, commuter and regional aircraft much less efficient than in the paper; the revised structure masses and fuel cell technology, the maximum power per propulsor and the later entry-into-service bounds further delay or remove the alternative architectures. No ablation study separates these contributions: the attribution is a reading of the design results (Part I), not a measured decomposition.

Part I – Models

The sections follow the paper's Methods (background scenarios, demand, fleet, energy, controls, optimization). For each, the table lists what is identical and what changes.

1. Background scenarios and demand

Unchanged. The AR6 SSP scenarios, the 2.0 °C carbon budget, the traffic demand (generalized logistic of GDP per capita) and the low-demand (demand aversion) formulation are those of the paper. Demand and its elasticity do not depend on the aircraft model, so the trend traffic is identical in both models; only the part of the demand that is supplied differs (Figure 8).

2. Fleet: aircraft design and technology

Changed – this is the update. The fleet logic (sigmoid adoption, market shares, retirement of the current fleet) is unchanged. The aircraft design model changes as follows:

- **GAM V3.0** replaces V2.0: refitted reference power and empty mass regressions, lift-to-drag regression distinguishing fan and propeller aircraft, refitted turboshaft consumption, and a scale factor for the efficiency loss of small turbofans.
- **Powertrain scale laws** (electric motor, power electronics, fuel cell) and a **maximum power per propulsor** available at entry into service. Battery-electric and fuel cell designs that cannot respect it are removed from a market (per-market feasibility filter), as the paper already does for electric aircraft.
- **Fuel cell**: system specific power without thermal management, plus a thermal management system (heat rejection per kilogram, parasitic power per heat), mission-averaged efficiency.
- **LH2 tanks**: the gravimetric index depends on the tank size and on the technology scenario.
- **Thermal engine**: efficiency trend with entry into service.
- **Technology scenarios**: four points (2020/2040/2060/2080) interpolated with PCHIP (`ScenarioTechParameter`) instead of three (2020/2040/2060).
- **Entry into service** lower bounds: Jet-A 2035, alternatives 2040; the initial entry into service of the LH2 gas turbine and the LH2 fuel cell is the same (2040). Turboprops are not part of the optimized fleets.
- **Constraints** added to the optimization for electric and fuel cell aircraft: unit power ratio ≤ 1, design closure gap ≤ 1e-4, relative efficiency gain ≥ 1.

Technology parameters of the Mid scenario (paper → update)

Parameter (Mid scenario)	Unit	2040 paper → update	2060 paper → update
Battery specific energy	Wh/kg	575 → 575	1050 → 1050
E-motor specific power (at 1 MW)	kW/kg	15.0 → 14.8	21.5 → 24.7
Power electronics specific power	kW/kg	20.0 → 20.0	26.0 → 26.0
Fuel cell specific power (update: without TMS)	kW/kg	2.50 → 2.00	4.50 → 2.75
Fuel cell system efficiency at cruise	%	50 → 46	58 → 49
Structure mass factor (relative to today)	%	78 → 90	70 → 86
E-motor efficiency (new)	%	→ 94.6	→ 96.3
Fuel cell TMS specific heat rejection (new)	kW/kg	→ 5.0	→ 8.0
Fuel cell TMS parasitic power per heat (new)	-	→ 0.17	→ 0.14
Maximum power per propulsor (new)	MW	→ 2.5	→ 4.0
Thermal engine efficiency factor (new)	%	→ 115	→ 125
LH2 tank mass factor k (new)	-	→ 2.84	→ 1.15
LH2 fuel system mass ratio s (new)	-	→ 0.262	→ 0.173

Energy per available seat-km of the new aircraft (Mid technology)

Energy per ASK of the new designs by entry into service (MJ per seat-km), compared with the 2019 fleet. 'infeasible' means the design does not close or violates the power limit; 'n/a' means the architecture is not available in the market in that model.

Market / architecture (Mid)	2019 fleet	2040 paper	2040 update	2060 paper	2060 update
general – Jet-A turbofan	1.91	0.39	1.61	0.37	1.44
general – LH2 gas turbine	1.91	0.47	2.33	0.44	1.80
general – LH2 fuel cell	1.91	0.48	1.26	0.36	0.69
general – Battery-electric	1.91	1.01	0.77	0.35	0.36
commuter – Jet-A turbofan	1.01	0.42	0.95	0.40	0.85
commuter – LH2 gas turbine	1.01	0.52	1.29	0.48	1.05
commuter – LH2 fuel cell	1.01	0.54	infeasible	0.40	0.63
commuter – Battery-electric	1.01	infeasible	infeasible	0.79	0.45
regional – Jet-A turbofan	0.80	0.52	0.80	0.49	0.71
regional – LH2 gas turbine	0.80	0.67	1.12	0.59	0.88
regional – LH2 fuel cell	0.80	0.71	infeasible	0.48	0.68
short medium – Jet-A turbofan	0.87	0.65	0.70	0.61	0.62
short medium – LH2 gas turbine	0.87	0.91	1.00	0.73	0.77
short medium – LH2 fuel cell	0.87	1.01	n/a	0.59	n/a
long range – Jet-A turbofan	0.92	0.85	0.80	0.79	0.68
long range – LH2 gas turbine	0.92	1.38	1.21	1.02	0.78
long range – LH2 fuel cell	0.92	1.59	n/a	0.79	n/a

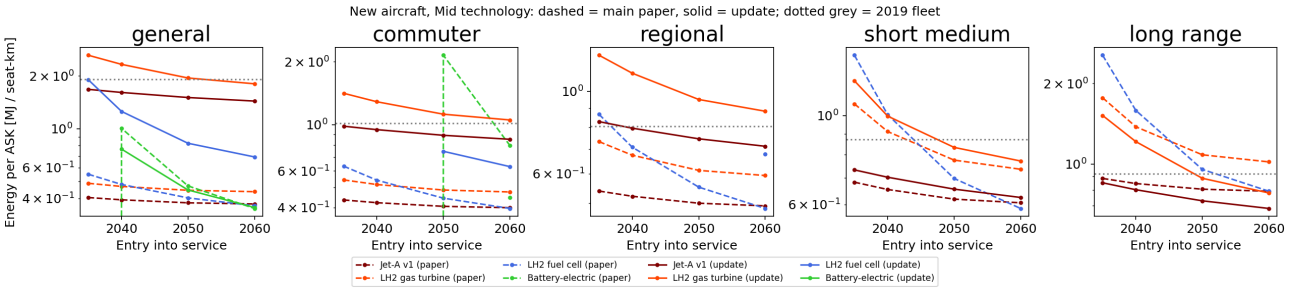


Figure 1 – Energy per ASK of the new aircraft versus entry into service, per market, Mid technology. Dashed: main paper; solid: update; dotted grey: 2019 fleet. Discontinuities show designs that do not close or violate the power limit. In the paper, some battery-electric designs have unphysical (negative or very large) values at early dates and are masked by the paper's own filter.

Reading: the update Jet-A aircraft are 2 to 4 times more energy consuming than the paper's in the general and commuter markets (small turbofans), 40–60 % in the regional market, and 2–8 % in the short-medium and long-range markets, where it is even better than the paper at long range. Fuel cell aircraft lose their advantage (system efficiency, thermal management, power limit) and are unavailable for short-medium and long-range. The LH2 gas turbine is hardly affected in the large markets and is penalised in the small ones.

3. Energy: carriers, pathways and primary energy

Unchanged. The energy carriers and production pathways (fossil, HEFA, ATJ, FT, electrolysis, gas reforming, power-to-liquid, hydrogen liquefaction, battery charging), their efficiencies and cost/resource impacts are those of the paper, in every technology scenario. The technology scenarios vary the aircraft technology only. **Added**: primary energy factors per carrier and per aircraft (documentation and figures only, no change in the optimization).

4. Controls and design space

Unchanged controls (aircraft market penetration, energy mix per pathway and period, supply cap for low demand). **Changed** bounds of the entry-into-service controls (paper: 2035–2060, 2030 for the first Jet-A generation; update: see Fleet) and the corresponding initial values, and the removal of the market/architecture combinations that are infeasible.

5. Optimization problem

Unchanged formulations (trend: minimise the cumulative CO2; low demand: minimise the time-discounted burden of demand aversion under the cumulative emissions budget), constraints on the biomass and electricity fair shares, solver (SLSQP through GEMSEO, automatic differentiation with JAX), tolerances and the 2000 (single) and 5000 (robust) iteration caps. **Added**: the three design constraints above, the differentiable design closure gap, the implicit adjoint through the Newton sizing solver, and checkpoint/warm start for the robust runs. All 29 optima

of the update (27 single, 2 robust) end feasible.

Part II – Results

The structure follows the paper's Results: baseline, mitigation (drop-in and breakthrough), robust policies, and the comparison with the literature.

6. Baseline scenarios (no mitigation measure)

Fossil-only scenarios. Without any alternative fuel or aircraft, only the fleet renewal with conventional aircraft reduces energy consumption; emissions in the update are higher by 25–30 % because the new Jet-A aircraft are less efficient, and the peak is later (2060 instead of 2041 in SSP1).

Baseline (Mid)	Cumulative CO2 [% budget]	Peak [x 2019]	Peak year	2075 [x 2019]
Baseline SSP1	220 → 276	1.57 → 2.14	2041 → 2060	1.49 → 2.04
Baseline SSP2	231 → 290	1.75 → 2.39	2075 → 2075	1.75 → 2.39
Baseline SSP5	317 → 398	2.30 → 3.14	2042 → 2061	2.26 → 3.08

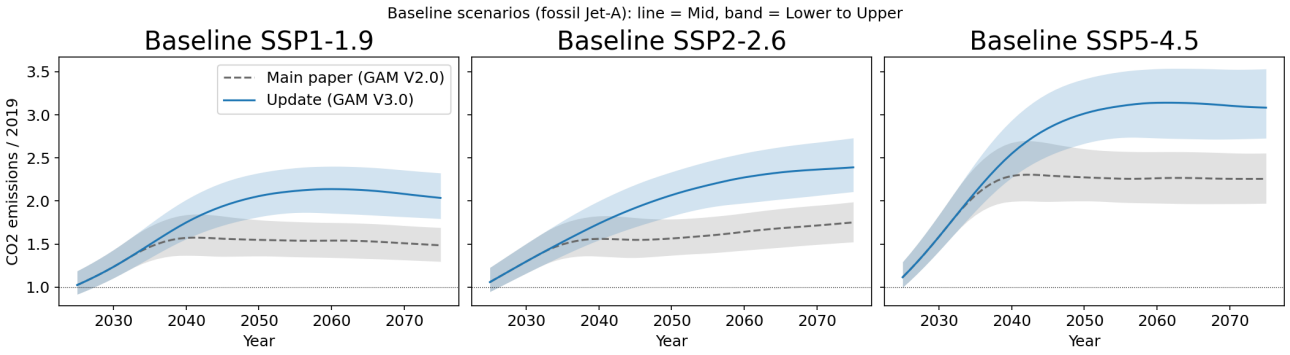


Figure 2 – Annual CO2 emissions of the baseline scenarios, Lower to Upper technology (bands), Mid line. Grey dashed: main paper; blue: update.

7. Mitigation scenarios

Headline metrics of the Mid technology optima (all variants). The update emits more in every case, and the ordering is preserved. Full results for the three technology levels are in Appendix C.

Scenario (Mid technology)	Cumulative CO2 [% budget]	Peak emissions [x 2019]	Peak year	Emissions 2075 [x 2019]	Avoided demand 2075 [%]	Fossil Jet-A 2075 [EJ]
Baseline SSP1	220 → 276	1.57 → 2.14	2041 → 2060	1.49 → 2.04	0 → 0	14.1 → 19.4
Baseline SSP2	231 → 290	1.75 → 2.39	2075 → 2075	1.75 → 2.39	0 → 0	16.7 → 22.8
Baseline SSP5	317 → 398	2.30 → 3.14	2042 → 2061	2.26 → 3.08	0 → 0	21.5 → 29.4
Drop-in trend	189 → 247	1.51 → 1.81	2040 → 2062	1.12 → 1.75	0 → 0	7.6 → 13.7
Drop-in availability	162 → 219	1.47 → 1.75	2040 → 2045	0.67 → 1.30	0 → 0	1.2 → 7.2
Drop-in low-demand	100 → 100	0.97 → 0.89	2036 → 2044	0.29 → 0.36	47 → 58	0.2 → 0.4
Breakthrough trend	166 → 216	1.52 → 1.77	2039 → 2044	0.69 → 1.01	0 → 0	3.6 → 6.4
Breakthrough availability	130 → 177	1.53 → 1.71	2039 → 2043	0.11 → 0.32	0 → 0	0.0 → 0.3
Breakthrough low-demand	100 → 100	1.13 → 1.02	2038 → 2042	0.08 → 0.13	39 → 45	0.0 → 0.1

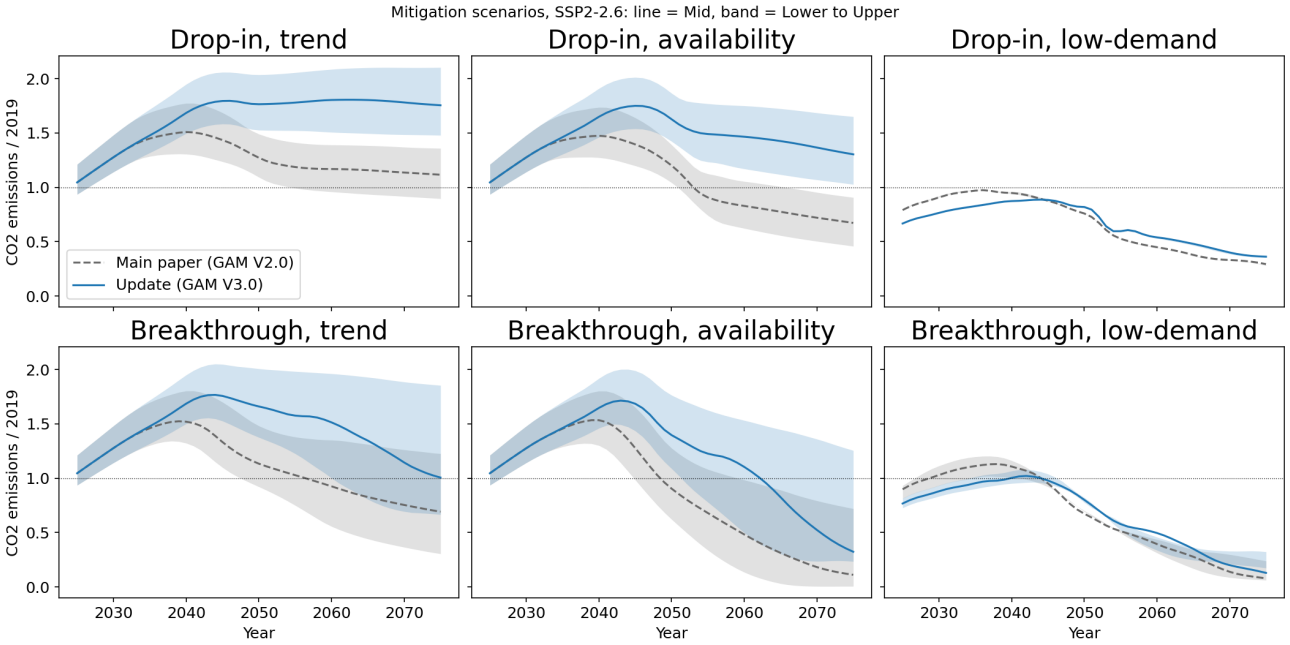


Figure 3 – Annual CO2 emissions of the drop-in and breakthrough scenarios (trend, availability, low demand) for the paper and the update.

Drop-in (only drop-in fuels, no alternative aircraft): the drop-in results are less affected by the aircraft design but still higher because the Jet-A new-generation aircraft are less efficient, so more biofuel and e-fuel are needed. **Breakthrough** (alternative architectures allowed): the update loses part of the benefit of hydrogen and electric aircraft, so the breakthrough advantage over drop-in is smaller (247 vs 216 % in the update, 189 vs 166 % in the paper).

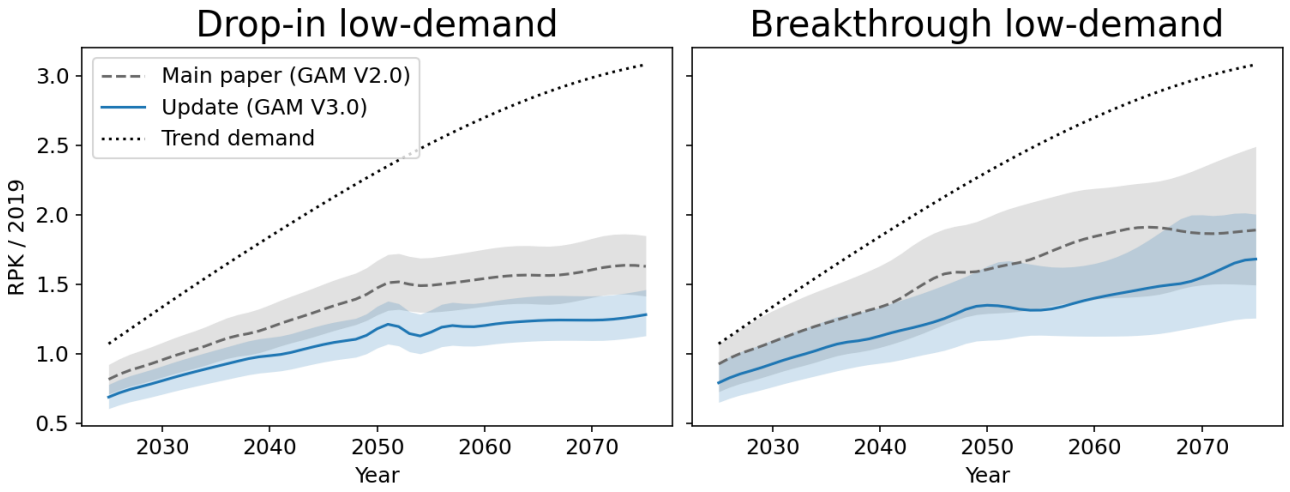


Figure 4 – Traffic of the low-demand variants relative to the trend (demand aversion).

Fleet mix

Share of supplied ASK in 2075 [%]	trend: paper → update	availability: paper → update	low-demand: paper → update
Lower – Current fleet	4 → 7	4 → 5	3 → 6
Lower – Jet-A v1	33 → 20	31 → 17	28 → 22
Lower – Jet-A v2	23 → 49	5 → 34	5 → 20
Lower – LH2 gas turbine	10 → 22	6 → 42	7 → 52
Lower – LH2 fuel cell	30 → 0	54 → 0	57 → 0
Lower – Battery-electric	0 → 2	0 → 2	0 → 0
Mid – Current fleet	1 → 5	1 → 2	2 → 4
Mid – Jet-A v1	10 → 16	5 → 9	7 → 6
Mid – Jet-A v2	29 → 30	4 → 13	2 → 9
Mid – LH2 gas turbine	1 → 0	2 → 15	2 → 18

Share of supplied ASK in 2075 [%]	trend: paper → update	availability: paper → update	low-demand: paper → update
Mid – LH2 fuel cell	57 → 29	85 → 43	84 → 44
Mid – Battery-electric	2 → 20	2 → 18	3 → 19
Upper – Current fleet	1 → 2	0 → 1	1 → 1
Upper – Jet-A v1	2 → 7	1 → 3	3 → 3
Upper – Jet-A v2	29 → 33	0 → 34	10 → 30
Upper – LH2 gas turbine	0 → 15	2 → 17	1 → 16
Upper – LH2 fuel cell	66 → 21	94 → 24	71 → 24
Upper – Battery-electric	3 → 23	3 → 22	15 → 26

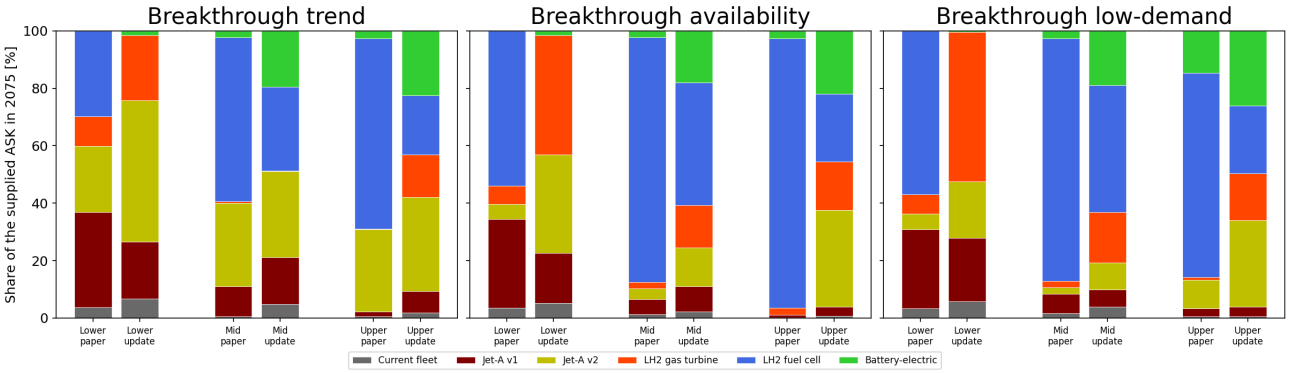


Figure 5 – Share of the supplied ASK in 2075 per architecture, Breakthrough scenarios.

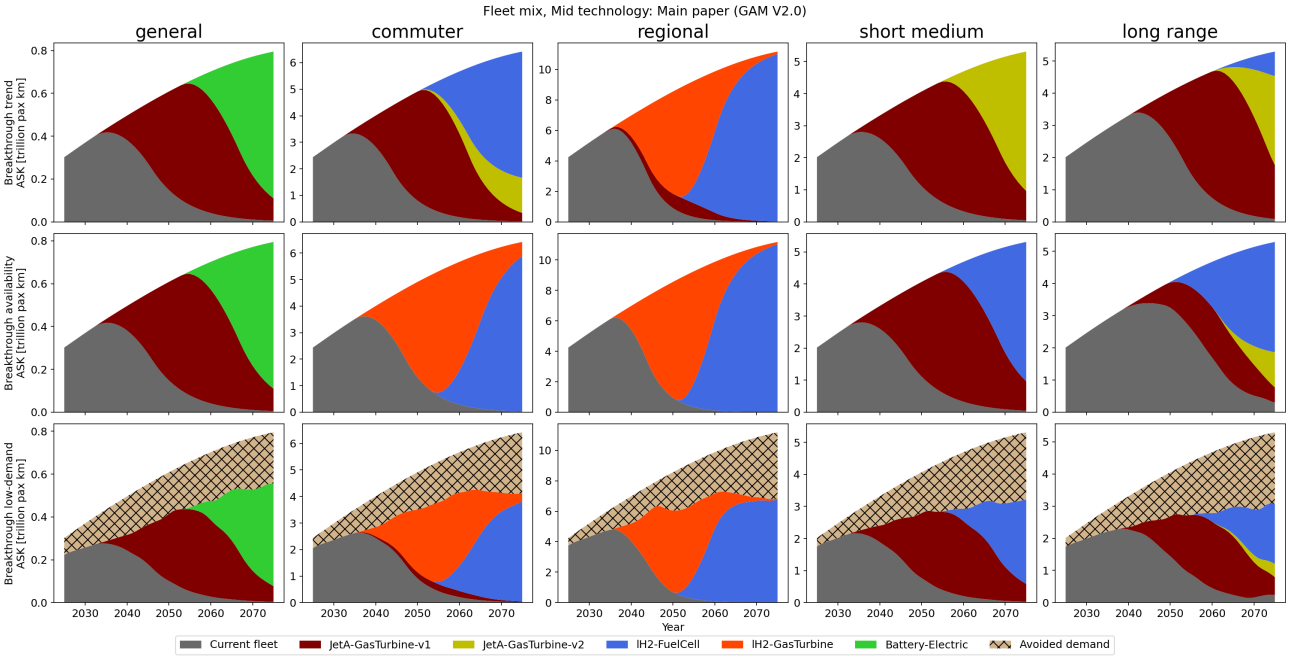


Figure 6 – Fleet mix per market, Mid technology, Breakthrough trend, main paper.

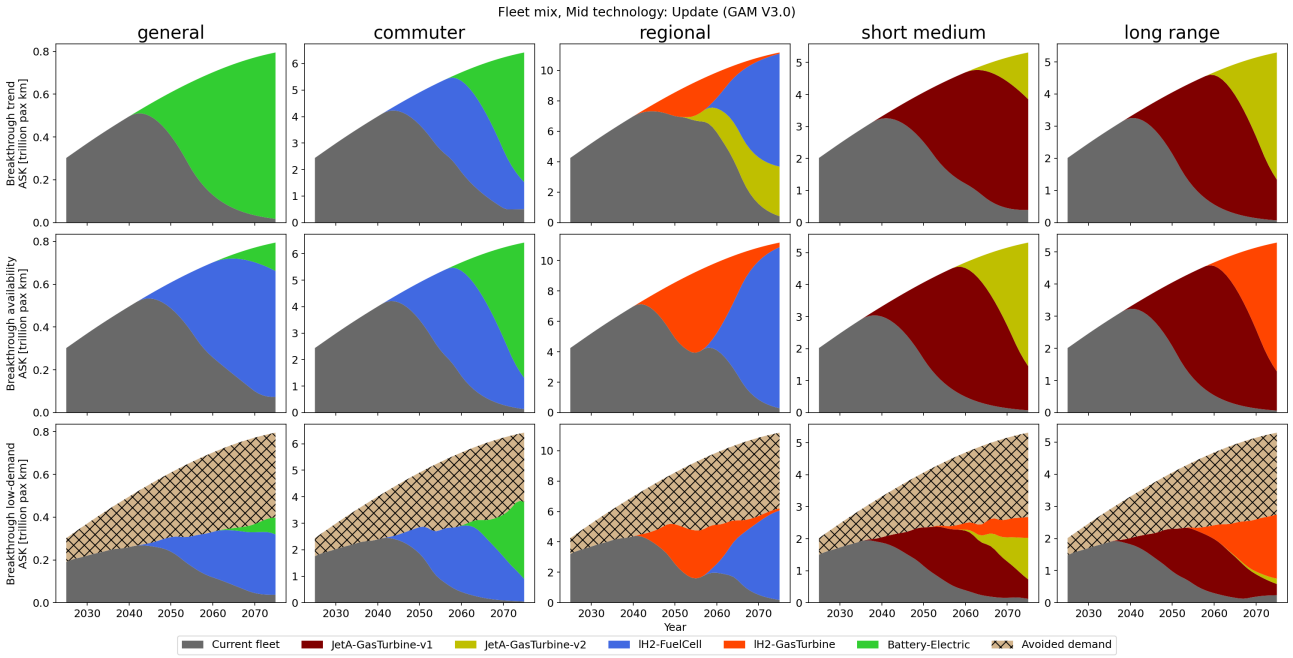


Figure 7 – Same, update. Jet-A v2, LH2 gas turbine and battery-electric aircraft partly replace the fuel cell aircraft; the 2019 fleet is renewed in every market.

Entry into service chosen by the optimizer, per architecture (range over the markets, number of markets with the architecture):

Mid technology, EIS range over the markets	Trend: paper	Trend: update	Availability: paper	Availability: update	Low-demand: paper	Low-demand: update
Jet-A v1	2032–2040 (5)	2035–2038 (5)	2032–2038 (5)	2035–2038 (5)	2032–2037 (5)	2035–2036 (5)
Jet-A v2	2050–2060 (5)	2051–2060 (5)	2052–2060 (5)	2051–2059 (5)	2052–2059 (5)	2056–2060 (5)
LH2 gas turbine	2035–2048 (5)	2040–2058 (5)	2035–2056 (5)	2040–2058 (5)	2035–2059 (5)	2040–2053 (5)
LH2 fuel cell	2050–2060 (5)	2041–2057 (3)	2049–2055 (5)	2041–2057 (3)	2050–2055 (5)	2041–2057 (3)
Battery-electric	2053–2059 (2)	2040–2057 (2)	2053–2059 (2)	2057–2060 (2)	2053–2059 (2)	2059–2060 (2)

Energy supply

Energy carriers produced and resources consumed (Mid, breakthrough). The update needs more fossil Jet-A (Part II table) because of the lower efficiency of the new fleet, while the biomass and electricity peaks are about the same because they are capped at their fair share of the global production.

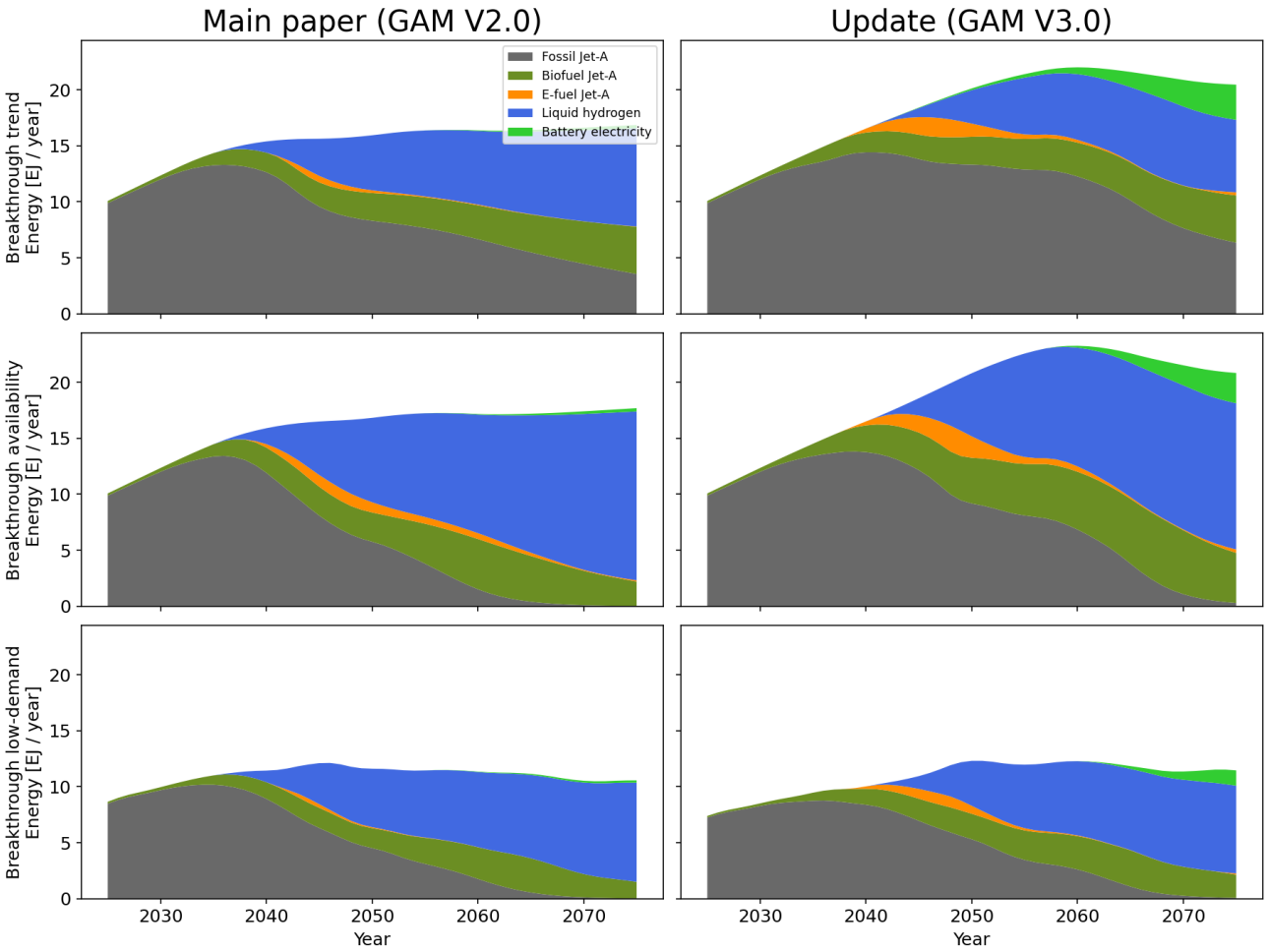


Figure 8 – Supply per final energy carrier, Mid technology, Breakthrough variants. Dashed: main paper; solid: update.

8. Robust policies

A single aircraft policy (the energy mix may adapt to each scenario) optimised on the mean objective over three SSP2 background scenarios (SSP2-19, SSP2-26, SSP2-34). In the update the trend policy no longer meets the budget in any of the three scenarios (it did not in the paper either), while the low-demand policy remains feasible.

Robust policy (Mid technology)	Scenario	Cumulative CO2 [% budget]	Peak [x 2019]	Emissions 2075 [x 2019]	Avoided demand 2075 [%]
Robust trend	SSP2-19	152 → 201	1.43 → 1.58	0.65 → 0.94	0 → 0
Robust trend	SSP2-26	167 → 216	1.52 → 1.76	0.70 → 1.00	0 → 0
Robust trend	SSP2-34	198 → 242	1.61 → 2.00	0.83 → 1.10	0 → 0
Robust low-demand	SSP2-19	100 → 100	1.15 → 1.02	0.06 → 0.12	40 → 46
Robust low-demand	SSP2-26	100 → 100	1.11 → 1.01	0.07 → 0.13	39 → 47
Robust low-demand	SSP2-34	100 → 100	0.92 → 0.91	0.04 → 0.08	46 → 55

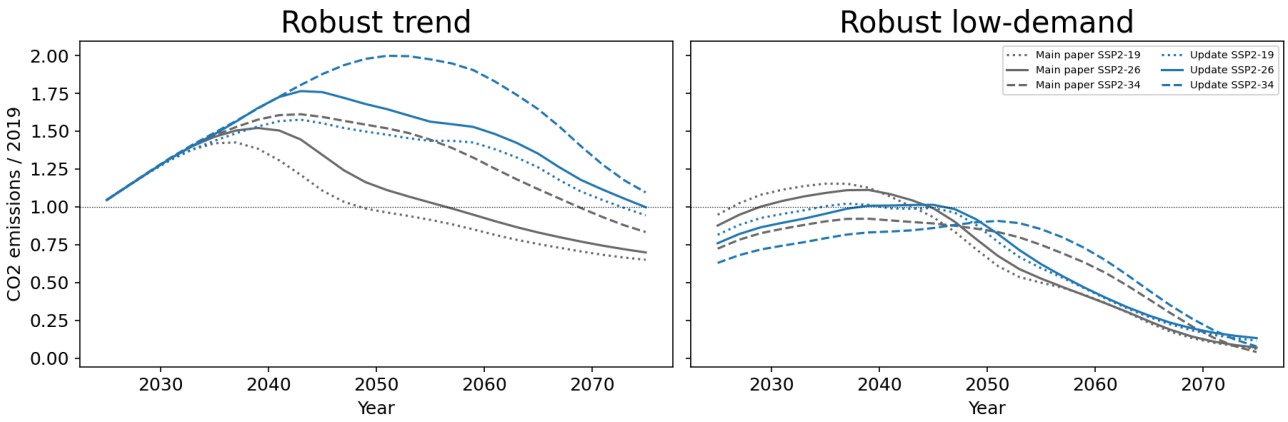


Figure 9 – Emissions of the robust policies in the three SSP2 variants.

9. Comparison with the literature

The paper compares its results with the literature (docs/paper/results/literature_comparison.md). That comparison is not recomputed here: the update emits more than the paper in every scenario, so the paper's position relative to the literature is expected to shift toward higher emissions, which would need to be checked with the same comparison script.

Discussion and limits

- **Attribution.** The increase of emissions comes from the combination of the model changes. The design table shows where the new Jet-A aircraft lose efficiency (small markets, GAM V3 turbofan scale factor) and where the alternative architectures are lost (power limit, fuel cell system). The share of each change in the emissions was not measured by a switch-by-switch study; running one would require optimizations with each update enabled alone.
- **Paper unchanged.** The paper model is the default and is guarded by `tests/test_gam_paper_baseline.py`; the paper results quoted here are the committed paper optima.
- **Optimization.** Optima are local (SLSQP); all update optima are feasible, but they were obtained from the same initial controls modified for the new bounds, so the paper-versus-update differences include optimizer effects that are small compared with the differences reported.
- **Not in this document.** The extended timeline (entry into service to 2080, end year 2100) is on hold, and geological hydrogen is planned work.

Appendix A – Where to find the results

All paths are relative to the repository root (`iancostalves/noads`, branch `ccr-906e3641-noua7t`).

What	Where
This document, its figures, scripts and tables	docs/update/comparison/ (update_vs_paper_comparison.pdf, figures/, results_summary.csv, build_pdf.py, comparison_data.py, comparison_figures.py, design_sweep.py, design_sweep.json, PR_DESCRIPTION.md)
Main paper optima (29)	docs/examples/optimization/results/<name>/opt_result.json
Update optima (29)	docs/examples/optimization/results/update/<name>/opt_result.json
Names	<scenario>-<tech> with tech = lowTech, midTech, upTech; scenarios SSP1-19-Fossil, SSP2-26-Fossil, SSP5-45-Fossil, SSP2-26-DropIn[-Availability|-LowDemand], SSP2-26[-Availability|-LowDemand]; robust-SSP2-midTech, robust-SSP2-lowdemand-LowDemand-midTech
Content of opt_result.json	'inputs' (optimal controls and constants such as entry into service) and 'outputs' (time series 2025–2075: CO2, cumulative.CO2, fleet ask per aircraft, energy production/consumption per carrier and pathway, rpk, rpk_trend, ...)
Update fleet-mix figures	docs/update/figures/fleet_mix_update_{lower,mid,upper}.png
Update model documentation	docs/update/index.md, aircraft_design.md, energy_pathways.md
Paper results documentation	docs/paper/results/ and docs/paper/models/
Code of the update	src/noads/gam_jax/models/gam_v3.py, src/noads/application/base_objects.py (parameters, feasibility filter), scenario_setup.py (entry-into-service bounds, constraints), examples.py (drivers, checkpoint)
Local only, not in Git	update/<name>/iterations.npz (every optimizer iteration), update/*.mp4 (iteration videos)

How the metrics are computed (`comparison_data.py`): cumulative CO2 is the last value of `cumulative.CO2` divided by 3 % of the 2.0 °C (66 %) budget; emissions are expressed relative to the 2019 value of 844.43 MtCO2; avoided demand is 1 – rpk/rpk_trend in 2075; architecture shares are the ASK of 2075 of the aircraft of each architecture over the total ASK of the market fleets.

Appendix B – Reproduction

```
pip install -e ".[test]" reportlab matplotlib
cd docs/examples/optimization/results
python precompute_update.py <name>  # recompute an update optimum (see the script for the list)
cd docs/update/comparison
rm design_sweep.json  # optional: recompute the design sweep
python build_pdf.py  # figures, results_summary.csv and this PDF
```

Appendix C – Full results, all technology levels

Paper → update, for every single-scenario optimum.

Scenario	Tech	Cum. CO2 [% budget]	Peak [x 2019]	Peak year	2075 [x 2019]	Avoided demand 2075 [%]	Fossil 2075 [EJ]	Biomass peak [EJ/yr]	Electricity peak [EJ/yr]
Baseline SSP1	Lower	253 → 313	1.85 → 2.40	2042 → 2060	1.69 → 2.32	0 → 0	16.1 → 22.1	0.0 → 0.0	0.0 → 0.0
Baseline SSP1	Mid	220 → 276	1.57 → 2.14	2041 → 2060	1.49 → 2.04	0 → 0	14.1 → 19.4	0.0 → 0.0	0.0 → 0.0
Baseline SSP1	Upper	193 → 243	1.37 → 1.87	2040 → 2056	1.30 → 1.79	0 → 0	12.4 → 17.1	0.0 → 0.0	0.0 → 0.0
Baseline SSP2	Lower	266 → 330	1.99 → 2.73	2075 → 2075	1.99 → 2.73	0 → 0	18.9 → 26.0	0.0 → 0.0	0.0 → 0.0
Baseline SSP2	Mid	231 → 290	1.75 → 2.39	2075 → 2075	1.75 → 2.39	0 → 0	16.7 → 22.8	0.0 → 0.0	0.0 → 0.0
Baseline SSP2	Upper	203 → 255	1.52 → 2.11	2075 → 2075	1.52 → 2.11	0 → 0	14.5 → 20.1	0.0 → 0.0	0.0 → 0.0
Baseline SSP5	Lower	364 → 454	2.70 → 3.54	2042 → 2062	2.55 → 3.53	0 → 0	24.3 → 33.6	0.0 → 0.0	0.0 → 0.0
Baseline SSP5	Mid	317 → 398	2.30 → 3.14	2042 → 2061	2.26 → 3.08	0 → 0	21.5 → 29.4	0.0 → 0.0	0.0 → 0.0
Baseline SSP5	Upper	278 → 351	2.00 → 2.74	2042 → 2057	1.97 → 2.73	0 → 0	18.8 → 26.0	0.0 → 0.0	0.0 → 0.0
Drop-in trend	Lower	223 → 287	1.77 → 2.10	2041 → 2075	1.36 → 2.10	0 → 0	9.9 → 17.0	7.1 → 7.1	13.4 → 13.4
Drop-in trend	Mid	189 → 247	1.51 → 1.81	2040 → 2062	1.12 → 1.75	0 → 0	7.6 → 13.7	7.2 → 7.1	13.4 → 13.4
Drop-in trend	Upper	161 → 213	1.31 → 1.58	2039 → 2045	0.90 → 1.48	0 → 0	5.5 → 11.1	7.1 → 7.1	13.4 → 13.4
Drop-in availability	Lower	196 → 259	1.74 → 2.01	2040 → 2045	0.91 → 1.65	0 → 0	3.4 → 10.5	12.3 → 12.3	23.0 → 23.0
Drop-in availability	Mid	162 → 219	1.47 → 1.75	2040 → 2045	0.67 → 1.30	0 → 0	1.2 → 7.2	12.3 → 12.3	23.0 → 23.0
Drop-in availability	Upper	135 → 186	1.28 → 1.54	2038 → 2045	0.46 → 1.03	0 → 0	0.2 → 4.6	12.3 → 12.3	23.0 → 23.0
Drop-in low-demand	Lower	100 → 100	0.98 → 0.89	2036 → 2043	0.28 → 0.36	54 → 63	0.2 → 0.4	7.1 → 7.2	13.4 → 13.4
Drop-in low-demand	Mid	100 → 100	0.97 → 0.89	2036 → 2044	0.29 → 0.36	47 → 58	0.2 → 0.4	7.1 → 7.1	13.4 → 13.4
Drop-in low-demand	Upper	100 → 100	0.98 → 0.89	2036 → 2044	0.29 → 0.36	40 → 53	0.2 → 0.4	7.1 → 7.1	13.4 → 13.4
Breakthrough trend	Lower	219 → 276	1.80 → 2.05	2041 → 2044	1.23 → 1.85	0 → 0	8.7 → 14.6	7.1 → 7.2	13.4 → 13.4
Breakthrough trend	Mid	166 → 216	1.52 → 1.77	2039 → 2044	0.69 → 1.01	0 → 0	3.6 → 6.4	7.1 → 7.1	13.4 → 13.3
Breakthrough trend	Upper	127 → 170	1.34 → 1.55	2038 → 2043	0.31 → 0.67	0 → 0	0.2 → 3.2	7.1 → 7.1	13.4 → 13.3
Breakthrough availability	Lower	188 → 240	1.80 → 2.00	2040 → 2044	0.72 → 1.25	0 → 0	1.7 → 6.8	12.3 → 12.3	23.0 → 23.0
Breakthrough availability	Mid	130 → 177	1.53 → 1.71	2039 → 2043	0.11 → 0.32	0 → 0	0.0 → 0.3	11.0 → 12.3	23.0 → 23.0
Breakthrough availability	Upper	94 → 132	1.32 → 1.50	2038 → 2042	0.01 → 0.23	0 → 0	0.0 → 0.1	8.3 → 12.3	19.6 → 23.0
Breakthrough low-demand	Lower	100 → 100	1.02 → 0.96	2037 → 2043	0.24 → 0.32	51 → 59	0.1 → 0.2	7.1 → 7.1	13.4 → 13.4
Breakthrough low-demand	Mid	100 → 100	1.13 → 1.02	2038 → 2042	0.08 → 0.13	39 → 45	0.0 → 0.1	6.7 → 7.1	13.4 → 13.4
Breakthrough low-demand	Upper	100 → 100	1.20 → 1.08	2037 → 2042	0.06 → 0.17	19 → 35	0.0 → 0.1	6.2 → 7.1	13.4 → 13.4